

Robin Erb

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EDUCATION

University of Copenhagen (KU)

Feb 2026 – July 2026

Semester Abroad (M.Sc. Computer Science)

Copenhagen, Denmark

– Relevant Coursework: Online and Reinforcement Learning, Web Recommender Systems, Search Engines

Heidelberg University

Apr 2025 – May 2027

Master of Science in Data and Computer Science | GPA: 1.3

Heidelberg, Germany

– Specialization: Machine Learning

– Relevant Coursework: Generative Neural Networks, Embedded Machine Learning, Machine Learning Essentials, Data Mining, Data Visualization

University of Konstanz

Oct 2021 – Mar 2025

Bachelor of Science in Computer Science | GPA: 1.4

Konstanz, Germany

– Specialization: Data Science and Human-Computer Interaction

– **Thesis (Grade: 1.0):** Developed a hybrid input device for 2D/3D interaction (Spatial Mouse [↗](#))

– Conducted a comprehensive user study and analytical evaluation using Python

Jacob-Grimm-Schule Kassel

Sep 2017 – Jun 2020

A Levels (Abitur) | GPA: 1.8

Kassel, Germany

PROFESSIONAL EXPERIENCE

Campus Five

Sep 2025 – Present

Freelance Software Developer

Remote

– Project: Automated Middleware and ERP Integration

- Built real-time middleware to sync sales, returns, and inventory between local ERP and Shopify
- Automated catalog management including product drafting, publishing, and archiving processes
- Engineered end-to-end system architecture using Google Cloud Functions and Firestore database

– Project: AI-Driven Sales Automation

- Implemented OCR data extraction for vehicle documents to support the automated sales pipeline
- Assisted in deploying an AI chatbot for inquiries and built a staff verification dashboard

University of Konstanz (HCI Department)

Sep 2023 – Mar 2025

Student Assistant

Konstanz, Germany

– Developed the "Spatial Mouse" prototype, including 3D CAD modeling

– Engineered Arduino-based hardware in C, involving circuit soldering and component assembly

– Built a dedicated virtual test environment for the prototype in Unity

Steinbock

Sep 2021 – Jul 2023

Bouldering Instructor (Part-time)

Konstanz, Germany

– Taught climbing techniques and safety protocols in youth bouldering classes

Kletterwerk Radolfzell

Sep 2020 – Aug 2021

Voluntary Service (Federal Volunteer Service - BFD)

Radolfzell, Germany

– Instructed school classes in bouldering and rope climbing

PUBLICATIONS

S. Hubenschmid, J. Zagermann, **R. Erb**, et al. "SpatialMouse: A Hybrid Pointing Device for Seamless Interaction Across 2D and 3D Spaces." In *Proc. of the 31st ACM Symposium on Virtual Reality Software and Technology (VRST '25)*. DOI: 10.1145/3756884.3766047

TECHNICAL SKILLS AND INTERESTS

Programming Languages: Python, Java, C++, SQL, R

Cloud & Tools: Google Cloud Functions, Firebase/SQL, Git, Docker, Unity, CAD

ML & Automation: OCR, AI-Bots, PyTorch, Scikit-learn, Pandas

Languages: German (Native), English (Fluent), French (Basic)

Interests: Rock Climbing, Volleyball, Snowboarding, DJing